

| Question |                  |                |  | Marking details                                                                                                                                                                                                                                              | Marks available |         |                |       |               |         |   |                  |         |  |   |  |   |  |  |
|----------|------------------|----------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------|----------------|-------|---------------|---------|---|------------------|---------|--|---|--|---|--|--|
|          |                  |                |  |                                                                                                                                                                                                                                                              | AO1             | AO2     | AO3            | Total | Maths         | Prac    |   |                  |         |  |   |  |   |  |  |
| 3        | (a)              |                |  | The alpha particle / it has <u>two</u> fewer [accept: less] electrons / no electrons [Not: just less / fewer electrons]<br>[Treat 'it' as the alpha particle]<br>Reference to charge neutral – not credited or penalised                                     | 1               |         |                | 1     |               |         |   |                  |         |  |   |  |   |  |  |
|          | (b)              | (i)            |  | <table><tr><td></td><td>Element</td><td>Nucleon number</td></tr><tr><td>X</td><td>Pb / lead (1)</td><td>214 (1)</td></tr><tr><td>Y</td><td>Bi / bismuth (1)</td><td>214 (1)</td></tr></table><br>NB. Additional numbers in the nucleon number box → no mark. |                 | Element | Nucleon number | X     | Pb / lead (1) | 214 (1) | Y | Bi / bismuth (1) | 214 (1) |  | 4 |  | 4 |  |  |
|          | Element          | Nucleon number |  |                                                                                                                                                                                                                                                              |                 |         |                |       |               |         |   |                  |         |  |   |  |   |  |  |
| X        | Pb / lead (1)    | 214 (1)        |  |                                                                                                                                                                                                                                                              |                 |         |                |       |               |         |   |                  |         |  |   |  |   |  |  |
| Y        | Bi / bismuth (1) | 214 (1)        |  |                                                                                                                                                                                                                                                              |                 |         |                |       |               |         |   |                  |         |  |   |  |   |  |  |
|          |                  | (ii)           |  | Z                                                                                                                                                                                                                                                            |                 | 1       |                | 1     |               |         |   |                  |         |  |   |  |   |  |  |
|          |                  |                |  | Question 3 total                                                                                                                                                                                                                                             | 1               | 5       | 0              | 6     | 0             | 0       |   |                  |         |  |   |  |   |  |  |